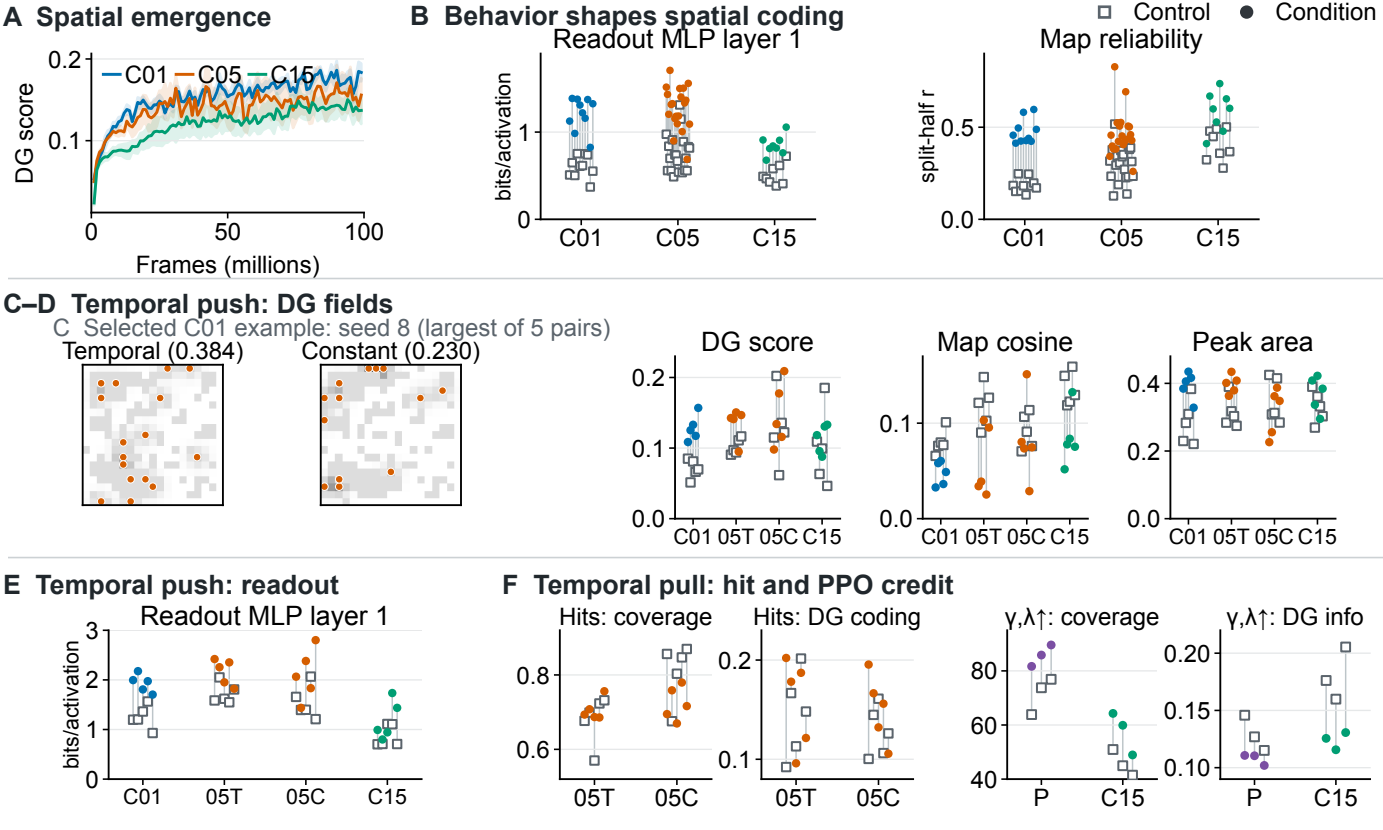

A Spatial emergence
C01
C05
C15
DG score
Frames (millions)
B Behavior shapes spatial coding
Control
Condition
Readout MLP layer 1
bits/activation
Map reliability
split-half r
C–D Temporal push: DG fields
C Selected C01 example: seed 8 (largest of 5 pairs)
Temporal (0.384)
Constant (0.230)
DG score
Map cosine
Peak area
C01 05T 05C C15
E Temporal push: readout
Readout MLP layer 1
F Temporal pull: hit and PPO credit
Hits: coverage
Hits: DG coding
γ,λ↑: coverage
γ,λ↑: DG info
05T
05C
P
C15